Inverse Analysis of Damped Oscillations

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Mini project for Differential Equations (Spring/2024)

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In this study, we explore the behavior of a damped spring-mass system through a two-part approach. Initially, we determine the general solution by solving the equations of motion and visualizing it graphically. Subsequently, we address an intriguing inverse problem by fitting experimental data to the theoretical model, thereby characterizing damping and angular frequency. By reconciling theory with observation, we are able to better understand the system properties based on our experimental findings.

Finding the general solution for Damped Harmonic Oscillator

Solving differential equation

A Mass - Spring System could be described by following equation

$$x''[t] + \gamma * x'[t] + \omega * x[t] = 0$$

Where

$$\omega^2 = \frac{k}{m}, \quad \gamma = \frac{b}{m}$$

```
oc[ω_, γ_] := x''[t] + γ * x'[t] + ω * x[t]
```

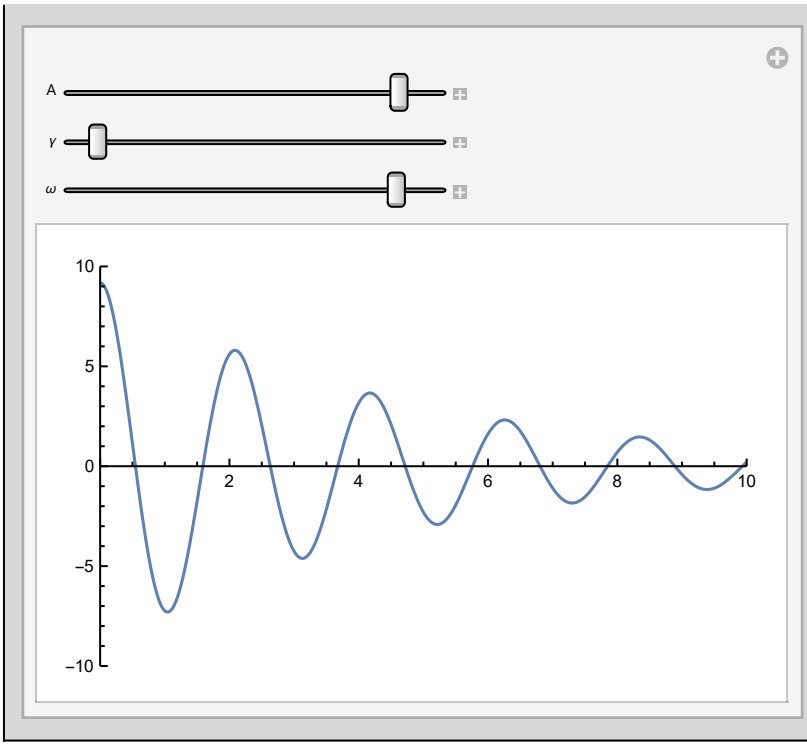
In[ ]:=
```
S = FullSimplify[DSolve[{oc[ω, γ] == 0, x[0] == A, x'[0] == 0}, x[t], t], {ω > 0, γ ≥ 0}]
```

Out[ ]=

$$\left\{\left\{x[t] \to A\, e^{-\frac{t\gamma}{2}} \left(\text{Cosh}\left[\frac{1}{2} t \sqrt{\gamma^2 - 4\omega}\right] + \frac{\gamma\, \text{Sinh}\left[\frac{1}{2} t \sqrt{\gamma^2 - 4\omega}\right]}{\sqrt{\gamma^2 - 4\omega}}\right)\right\}\right\}$$

In[ ]:=
```
Manipulate[Plot[A e^(-t γ/2) (Cosh[1/2 t Sqrt[γ^2 - 4 ω]] + γ Sinh[1/2 t Sqrt[γ^2 - 4 ω]] / Sqrt[γ^2 - 4 ω]), {t, -10, 10},
  PlotRange → {{0, 10}, {-10, 10}}], {A, 0, 10}, {γ, 0, 10}, {ω, 0.01, 10}]
```

Out[ ]=



Simple Harmonic Motion

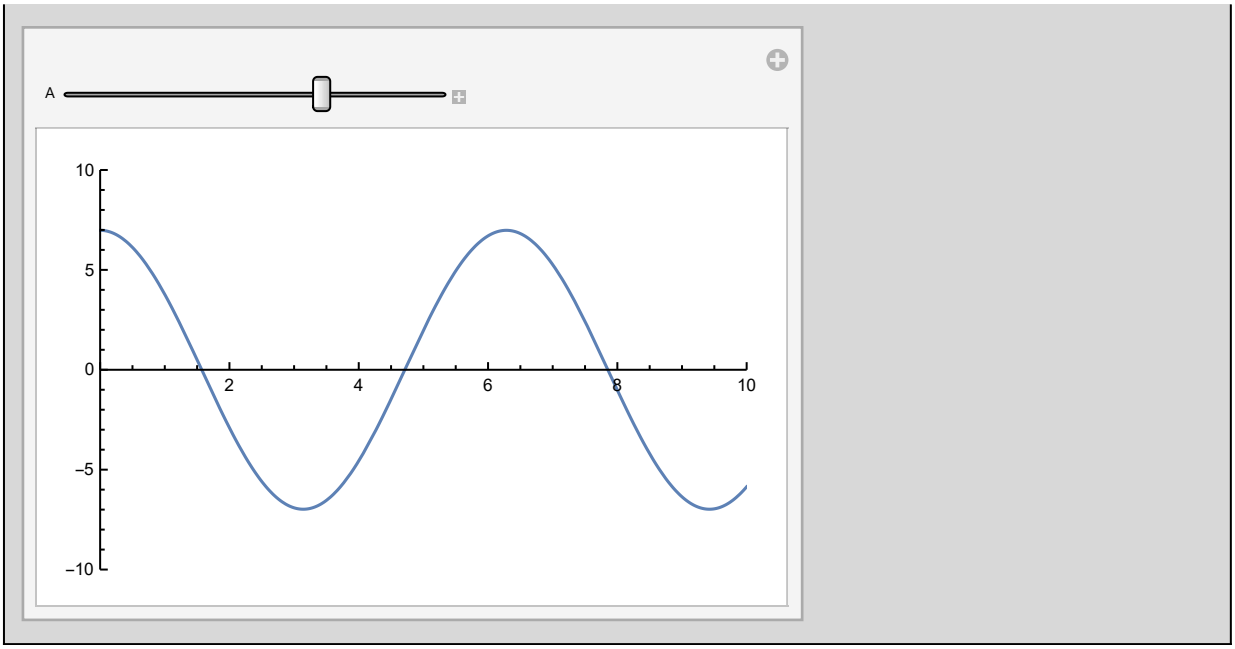
Assuming γ=0 we have a Simple Harmonic Motion

In[ ]:=
```
S1 = FullSimplify[DSolve[{oc[1, 0] == 0, x[0] == A, x'[0] == 0}, x[t], t], {ω > 0, γ ≥ 0}]
```

Out[ ]=
```
{{x[t] → A Cos[t]}}
```

In[ ]:=
```
Manipulate[Plot[A Cos[t], {t, -10, 10}, PlotRange → {{0, 10}, {-10, 10}}], {A, 0, 10}]
```

Out[ ]=



Damped Harmonic Oscillator

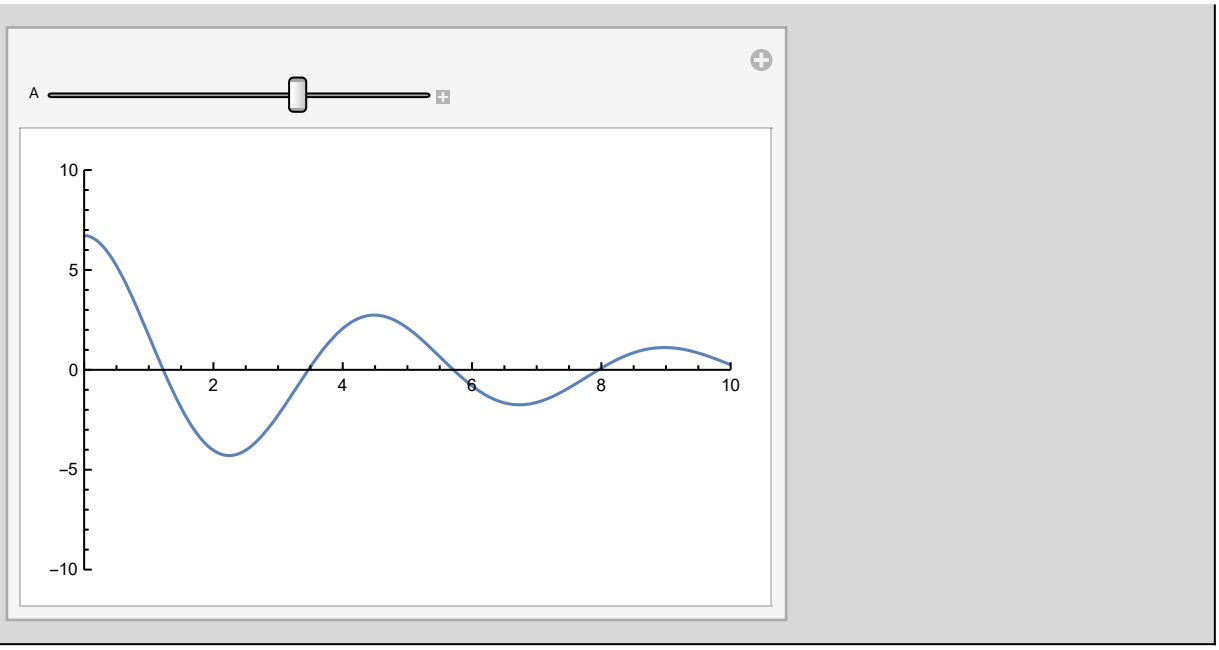
In[ ]:=
```
S2 = DSolve[{oc[2, .4] == 0, x[0] == A, x'[0] == 0}, x[t], t]
```

Out[ ]=

$$\left\{\left\{x[t] \to 1.\, e^{-0.2\,t}\,(1.\, A \operatorname{Cos}[1.4\,t] + 0.142857\, A \operatorname{Sin}[1.4\,t])\right\}\right\}$$

In[ ]:=
```
Manipulate[Plot[1.` e^(-0.2` t) (1.` A Cos[1.4` t] + 0.14285714285714288` A Sin[1.4` t]),
  {t, -10, 10}, PlotRange → {{0, 10}, {-10, 10}}], {A, 0, 10}]
```

Out[ ]=



Inverse Problem

Generating Data

in this part we are going to make a data list like this :

list = $\{\{t_1, x[t_1]\}, \{t_2, x[t_2]\}, \ldots, \{t_n, x[t_n]\}\}$

In[ ]:=
```
tmax = 20;
time = Range[0, tmax, 2 π / tmax];
```

In[ ]:=
```
DSolve[{oc[1, 0] == 0, x[0] == 1, x'[0] == 0}, x[t], t]
```

Out[ ]=
```
{{x[t] → Cos[t]}}
```

In[ ]:=
```
data1 = Table[{i, Cos[i]}, {i, time}];
```

In[ ]:=
```
DSolve[{oc[1, 0.3] == 0, x[0] == 3, x'[0] == 0}, x[t], t]
```

Out[ ]=
$$\{\{x[t] \to 3.\, e^{-0.15\, t}\, (1.\, \text{Cos}[0.988686\, t] + 0.151717\, \text{Sin}[0.988686\, t])\}\}$$

In[ ]:=
```
data2 = Table[{i, 3.` ℯ^(-0.15` i) (1.` Cos[0.9886859966642595` i] +
        0.15171652122725207` Sin[0.9886859966642595` i])}, {i, time}];
```

In[ ]:=
```
DSolve[{oc[0.5, 1.6] == 0, x[0] == 3, x'[0] == 0}, x[t], t]
```

Out[ ]=
$$\{\{x[t] \to 4.70713\, e^{-1.6\, t}\, (-0.36267\, e^{0.425834\, t} + 1.\, e^{1.17417\, t})\}\}$$

In[ ]:=
```
data3 = Table[{i, 4.707134902949092` ℯ^(-1.6` i)
      (-0.36266963623233867` ℯ^(0.4258342613226058` i) + 1.` ℯ^(1.1741657386773943` i))}, {i, time}];
```

Fitting data to the solution

Now we can fit our data to the general solution and find Amplitude, Angular Frequency and Damping index

In[ ]:=
```
FindFit[data1, A * e^(-t γ/2) (Cosh[1/2 t Sqrt[γ^2 - 4 ω]] + γ Sinh[1/2 t Sqrt[γ^2 - 4 ω]] / Sqrt[γ^2 - 4 ω]), {A, γ, ω}, t]
```

Out[ ]=
$$\{A \to 1.,\ \gamma \to -1.84575 \times 10^{-15},\ \omega \to 1.\}$$

In[ ]:=
```
FindFit[data2, A * e^(-t γ/2) (Cosh[1/2 t Sqrt[γ^2 - 4 ω]] + γ Sinh[1/2 t Sqrt[γ^2 - 4 ω]] / Sqrt[γ^2 - 4 ω]), {A, γ, ω}, t]
```

Out[ ]=
$$\{A \to 3.,\ \gamma \to 0.3,\ \omega \to 1.\}$$

In[ ]:=
```
FindFit[data3, A * e^(-t γ/2) (Cosh[1/2 t Sqrt[γ^2 - 4 ω]] + γ Sinh[1/2 t Sqrt[γ^2 - 4 ω]] / Sqrt[γ^2 - 4 ω]), {A, γ, ω}, t]
```

Out[ ]=
$$\{A \to 3.,\ \gamma \to 1.6,\ \omega \to 0.5\}$$